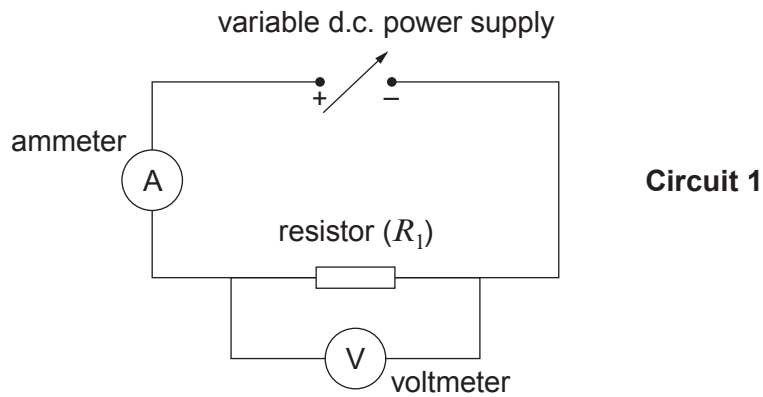


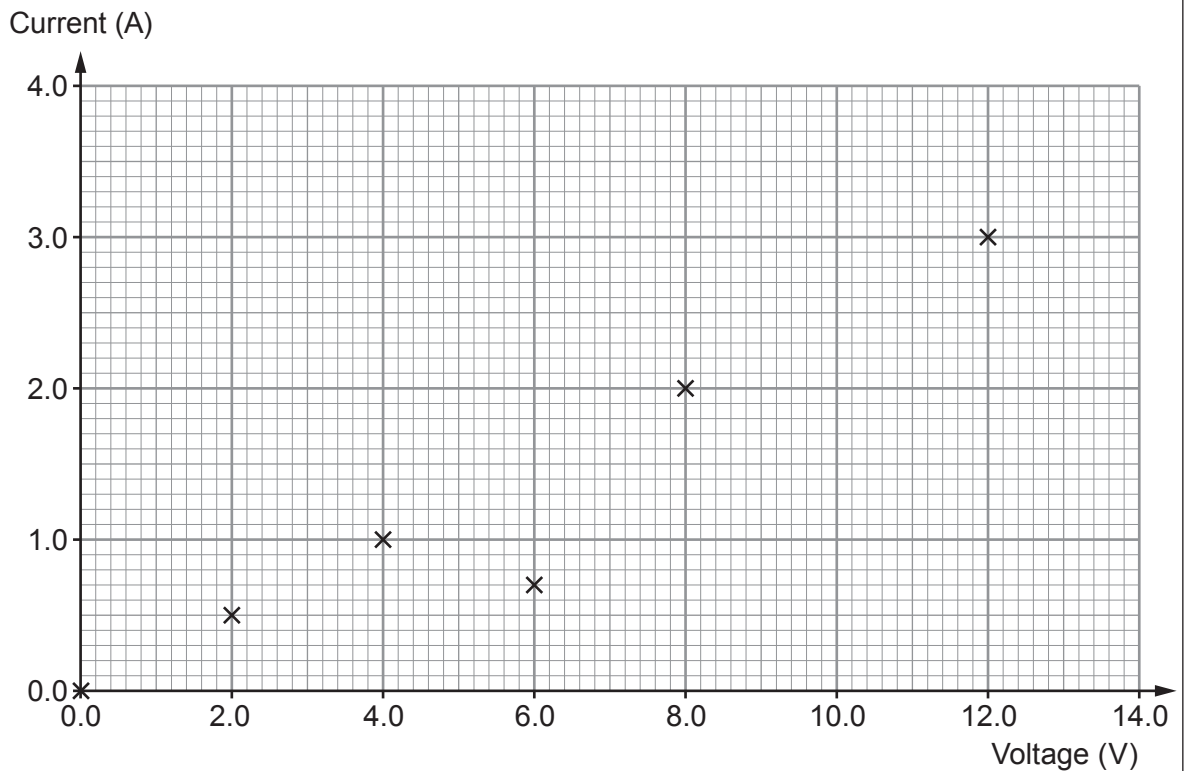
Answer **all** questions.

1. Paul investigated how the current changes with voltage for a resistor (R_1). The circuit he used is shown (**Circuit 1**).



- (a) Name the device he used to measure the current. [1]

- (b) Paul's results are shown on the graph.



- (i) On the graph, **circle** the anomalous point. [1]

- (ii) **Complete the graph** by drawing a suitable straight line. [1]



(iii) I. Use the graph to state the current at 12.0 V. [1]

Current = A

II. Use the equation:

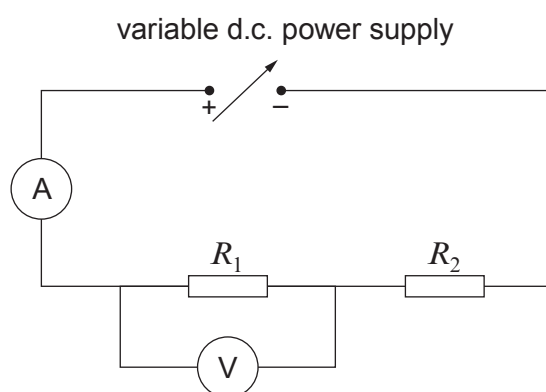
$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the resistor (R_1) at 12.0 V. [2]

Resistance of R_1 = Ω

(iv) State what the graph shows about the resistance of the resistor (R_1) as the voltage changes. [1]

- (c) The original circuit is then adapted by adding a second, **identical** resistor (R_2) in series with R_1 .
This circuit is shown below (**Circuit 2**).



Circuit 2

Complete the following sentences by underlining the correct word or phrase in each bracket. [3]

The total resistance in circuit 2 is (**double / the same as / half**) the total resistance of circuit 1.

When the d.c. power supply is set at the same voltage, the voltmeter reading in circuit 2 is (**double / the same as / half**) the voltmeter reading in circuit 1.

The current in circuit 2 is (**double / the same as / half**) the current in circuit 1.



- (b) (i) A salesperson suggests to the farmer that **wind generator 3** is better than **wind generator 2** because it is cheaper.
Explain, using information from the table, if the salesperson's suggestion is correct over a 5-year timescale. [2]

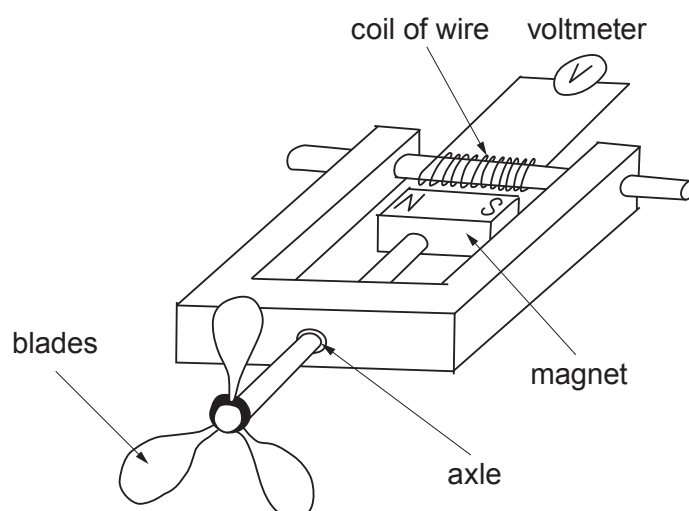
.....

.....

.....

- (ii) Give **two** reasons why the payback time can only be an estimate. [2]
1.
2.

- (c) A wind generator only operates in certain wind speeds.
The diagram below shows a student's design for measuring wind speed.



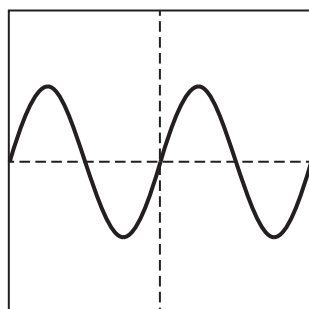
- (i) When the wind blows a reading on the voltmeter is obtained.
The following sentences describe the different stages.
- A** The magnetic field lines cut through the coil of wire.
 - B** The blades rotate.
 - C** The magnet rotates.
 - D** This induces a voltage in the coil of wire.

Put the letters **A**, **B** and **C** in the boxes to correctly describe the process of inducing a voltage in the coil of wire. The final box has already been completed, [2]

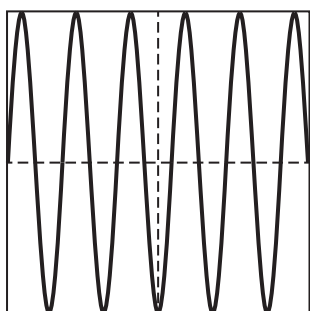
→ → → **D**



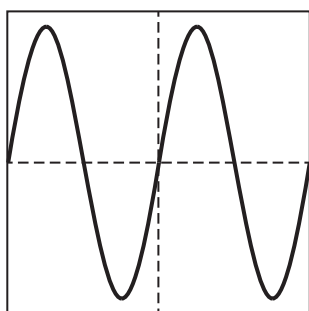
- (ii) When the wind speed is **2 m/s** the alternating voltage produced is shown below.



The following traces **X**, **Y** and **Z** are produced during different wind conditions.



Trace **X**



Trace **Y**



Trace **Z**

- I. State which trace (**X**, **Y** or **Z**) is produced when the wind speed is **faster than 2 m/s**.

Trace [1]

- II. State which trace (**X**, **Y** or **Z**) is produced when **the wind stops**.

Trace [1]

- (d) When the student's design is built it is found that the voltmeter is not sensitive enough to measure a wind speed less than 0.5 m/s.

- (i) Suggest **one** change that can be made to the **magnet** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

.....

- (ii) Suggest **one** change that can be made to the **coil of wire** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

.....



(b) Household circuits may be protected by a series of devices. These include:

- a miniature circuit breaker (mcb)
- a fuse in a plug
- a residual current circuit breaker (rccb).

The list below gives three situations in which one or another of these devices stop the power from being supplied.

1. An electric lawn mower cuts through the cable.
2. Too much current flows in a ring main because too many appliances are being used at the same time.
3. The live wire in a kettle touches its neutral wire.

(i) Select and explain one situation above that puts the **user** in danger of an electric shock **and** which one of the devices should be used to keep the user safe. [2]

.....

.....

.....

(ii) State **two** advantages of an mcb over a fuse in a household circuit. [2]

1.
2.

(c) Describe the function of the live and neutral wires in household circuits. [2]

Live wire:

.....

Neutral wire:

.....

